

HANDWRITTEN NUMBER RECOGNITION

Using Convolutional Neural Networks

Machine Learning Mini Project

Recognizes handwritten single-digit and multi-digit numbers from real-world photographs using OpenCV and a CNN.

99.21% MNIST Test Accuracy	0–9 Digit Classes	Single + Multi Number Recognition
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This document explains the project in simple terms: what it uses, how it works, how to run it, and what the application can recognize.

1. Technologies Used

Technology	Purpose
Python	Main programming language
TensorFlow / Keras	Build and train the CNN
OpenCV	Process images and find digits
NumPy	Work with image data
Pillow	Load and display images
CustomTkinter	Create the desktop application
Matplotlib	Testing and visualizations

2. Project Structure

```
HandwrittenDigitRecognition/  
  ■■■ model/digit_model.keras  
  ■■■ test_images/  
  ■■■ requirements.txt  
  ■■■ train_model.py  
  ■■■ predict.py  
  ■■■ app.py  
  ■■■ README.md
```

train_model.py — trains the CNN using MNIST.

predict.py — tests predictions from images.

app.py — runs the desktop application.

digit_model.keras — stores the trained model.

test_images/ — contains sample handwritten images.

3. How the System Works

The complete process is simple:



4. CNN Model

The project uses a Convolutional Neural Network (CNN) trained on the MNIST handwritten digit dataset.

Input 28×28 → Convolution → Pooling → Convolution → Pooling → Dense → Output 0-9

Data augmentation: During training, images are slightly rotated, moved, and zoomed. This helps the CNN handle real handwritten images better.

MNIST test accuracy: approximately 99.21%.

5. Image Processing

Step	What happens
1	Convert the image to grayscale.
2	Separate the handwriting from the background.
3	Find the handwritten digit areas.
4	Remove small unwanted noise.
5	Sort multiple digits from left to right.
6	Convert each digit into a 28×28 image.
7	Send each digit to the CNN.
8	Join the predictions to form the final number.

6. Single & Multiple Digit Recognition

The application supports both one digit and complete numbers.

Input	Output
7	7
42	42
019724	019724

The result is stored as text, so leading zeros are preserved. For example, 019724 stays 019724.

7. How to Run the Project

Install Python 3.10 or 3.11 and make sure Python is added to PATH.

Open PowerShell in the project folder:

```
cd C:\Users\\Desktop\HandwrittenDigitRecognition
```

Create and activate the virtual environment:

```
python -m venv venv
.\venv\Scripts\Activate.ps1
```

Install the required libraries:

```
pip install -r requirements.txt
```

8. Train the Model

The trained model is already included, so training again is optional.

```
python train_model.py
```

9. Test Predictions

Test all sample images:

```
python predict.py --all
```

Test one image:

```
python predict.py test_images/7.jpg
```

10. Run the Desktop Application

Start the application:

```
python app.py
```

HOW TO USE

1. Click **Select Image**.
2. Choose a handwritten number image.
3. Click **Predict Number**.
4. View the predicted number and confidence.
5. Click **Clear** to test another image.

11. Final Result

The project combines **Python + OpenCV + CNN + MNIST + CustomTkinter** to recognize handwritten single-digit and multi-digit numbers from photographs.

Example	Result
Handwritten 7	7
Handwritten 42	42
Handwritten 019724	019724